CPTG: 16/50 Stable Structural Floor

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Abstract

In Curvature Polarization Transport Gravity, the 16/50 shell diagnostic represents a stable structural floor. It describes the point where a galaxy's internal curvature organization and the surrounding geometric expansion scale enter balance. This ratio also connects the CPTG geometric expansion scale to the standard matter-based expansion scale.

1 Shell Diagnostic

The shell diagnostic is

$$\frac{16}{50} = 0.32.$$

In CPTG, this value acts as a structural equilibrium point. It is the ratio where the baryonic core and the surrounding geometric curvature response reach a stable balance.

This can be written as

$$H_{0,\text{standard}} \approx \frac{16}{50} H_{0,\text{CPTG}}.$$

Using

$$H_{0,\text{standard}} \approx 67 \text{ km/s/Mpc}$$

and

$$H_{0,\text{CPTG}} \approx 207 \text{ km/s/Mpc},$$

the ratio is

$$\frac{67}{207} \approx 0.323.$$

This places the observed matter-based expansion scale very close to the 16/50 CPTG shell.

2 Cooling into the Shell

At very high redshift, such as

$$z \approx 14,$$

a forming galaxy is in a dense, high-acceleration state. Its nonlinear curvature response is regulated by the suppression governor,

$$\Sigma = \left[1 + \left(\frac{g}{a_\star}\right)^{123/50}\right]^{-11/50}.$$

As the universe expands, local density decreases and the internal structure of the galaxy settles.

The structural mode N moves from an early high-energy organization toward a more stable configuration. In this process, the galaxy can migrate toward the 16/50 shell.

The transition is

$$\text{early high-density state} \quad \rightarrow \quad \text{suppressed curvature response} \quad \rightarrow \quad \text{stable 16/50 shell.}$$

Once the system reaches this shell, it becomes a long-lived bound structure.

3 Baryonic Core and Geometric Halo

The 16/50 shell describes a balance between two parts of the CPTG galaxy system:

$$\text{baryonic core}$$

and

$$\text{geometric curvature response.}$$

The baryonic core supplies the ordinary matter source. The geometric curvature response supplies the surrounding nonlinear structure.

At the shell floor,

$$\frac{\text{settled expansion response}}{\text{full geometric expansion response}} \approx \frac{16}{50}.$$

This means the galaxy is not using the full unsuppressed curvature expansion scale. It has settled into a stable fraction of it.

4 Link to the Hubble Tension

The same ratio appears in the comparison between the standard expansion scale and the CPTG geometric expansion scale:

$$\frac{67}{207} \approx 0.323 \approx \frac{16}{50}.$$

In CPTG, this reframes the Hubble tension.

The standard value near

$$67 \text{ km/s/Mpc}$$

represents the settled matter-based expansion scale.

The CPTG value near

$$207 \text{ km/s/Mpc}$$

represents the full structured geometric expansion scale.

Thus, the gap is not simply a disagreement between measurements. It is interpreted as the difference between the shell floor and the full geometric ceiling:

$$\text{floor} = \frac{16}{50},$$

$$\text{ceiling} = \frac{50}{50}.$$

5 Prediction

If the 16/50 shell is structural, stable galaxies should not settle into random final ratios.

The CPTG prediction is

$$\text{stable non-merging galaxies} \quad \to \quad \frac{16}{50} \text{ shell behavior.}$$

This should hold across redshift once a galaxy has reached a stable internal configuration.

Galaxies at different epochs may be observed at different stages of this process, but mature non-merging systems should tend back toward the same shell diagnostic.

6 Redshift Test

The shell model can be tested by comparing stable galaxies across redshift:

$$z = 0,\ 2,\ 5,\ 10,\ 14.$$

If galaxies settle randomly, the 16/50 interpretation fails.

If stable systems repeatedly return toward

$$0.32,$$

then the shell diagnostic is supported.

The useful test is not whether every early galaxy is already settled. The useful test is whether stable systems show convergence toward the same structural floor as they mature.

7 Universal Anchor

The 16/50 shell acts as an anchor in CPTG.

It connects

$$\text{early curvature transport,}$$

$$\text{cosmological expansion scale,}$$

and

$$\text{galaxy structural settling.}$$

In this view, the 16/50 shell is not only a galaxy-scale diagnostic. It is a structural ratio connecting the early lattice, the Hubble scale, and the long-lived organization of galaxies.

8 Summary

The 16/50 shell diagnostic is

$$\frac{16}{50} = 0.32.$$

It represents the stable floor where galaxy structure settles into balance with the surrounding geometric curvature response.

The same ratio appears in the expansion-scale comparison:

$$\frac{67}{207} \approx 0.323 \approx \frac{16}{50}.$$

CPTG therefore interprets the Hubble tension as a structural ratio between the observed matter-based expansion scale and the full geometric expansion scale.

In compact form:

$$H_{0,\text{standard}} \approx \frac{16}{50} H_{0,\text{CPTG}}.$$

The 16/50 shell is the universal anchor linking early curvature structure, cosmic expansion, and stable galaxy organization.